Supplementary Material to "Logic-Based Benders decomposition for additive manufacturing scheduling with sequence-dependent setup times" by Wenjing Ma, Jian Chen, Yaowen Sang, Yingqian Zhang, Yaoxin Wu

S.1 Detailed Computational Results

In the following, we report the detailed results from our computational study.

Table 1 Detailed results of three MIP models

Instances	MIP1			MIP2			MIP3		
	Obj	Gap	Time	Obj	Gap	Time	Obj	Gap	Time
I2_R10_M2-1	154.92	60.06	1800.15	154.92	0	32.22	154.92	35.55	1800.12
I2_R10_M2-2	186.93	66.67	1800.30	186.93	0	10.37	186.93	33.98	1800.63
I2_R10_M2-3	287.87	71.50	1800.35	287.88	0	15.49	287.88	59.63	1800.63
I2_R10_M2-4	203.31	66.40	1800.34	203.31	0	2.96	203.31	0	409.11
I2_R10_M2-5	196.44	72.52	1800.26	196.44	0	5.96	196.44	32.18	1800.11
I2_R10_M3-1	418.75	58.62	1800.24	418.75	0	37.24	418.75	59.37	1800.30
I2_R10_M3-2	373.08	73.31	1801.49	373.08	0	28.68	373.07	59.22	1800.51
I2_R10_M3-3	309.66	37.32	1800.67	309.66	19.05	1800.13	309.66	35.40	1800.40
I2_R10_M3-4	260.23	56.70	1800.36	260.23	0	333.21	260.23	54.90	1800.71
I2_R10_M3-5	299.98	49.95	1800.55	299.98	0	566.98	299.98	47.97	1800.87
I2_R20_M2-1	234.83	72.71	1800.45	230.00	60.66	1800.34	230.00	71.47	1800.28
I2_R20_M2-2	317.36	82.11	1800.99	271.35	78.45	1800.46	271.35	78.45	1801.67

I2_R20_M2-3	397.81	77.74	1800.92	373.05	75.78	1800.50	372.45	78.45	1801.39
I2_R20_M2-4	288.13	75.07	1801.24	273.70	51.50	1800.57	271.41	84.84	1802.04
I2_R20_M2-5	308.53	81.50	1800.26	293.65	69.71	1800.57	293.65	94.60	1801.59
I2_R20_M3-1	295.72	83.41	1801.30	284.99	82.35	1800.61	284.42	92.46	1802.29
I2_R20_M3-2	306.60	79.00	1801.30	302.93	78.26	1800.50	302.93	99.42	1801.33
I2_R20_M3-3	361.19	80.38	1801.01	318.73	77.19	1800.50	318.31	77.94	1801.03
I2_R20_M3-4	347.66	82.27	1801.11	336.69	81.30	1800.51	339.71	82.48	1801.10
I2_R20_M3-5	347.95	69.31	1801.06	345.98	68.76	1800.52	356.00	69.67	1801.56
I5_R10_M2-1	72.69	16.08	1800.31	72.69	0	4.05	72.69	0	74.31
I5_R10_M2-2	84.86	26.54	1800.36	84.86	0	407.59	84.86	30.53	1800.23
I5_R10_M2-3	56.98	30.22	1800.51	56.98	0	174.33	56.98	33.21	1801.13
I5_R10_M2-4	77.65	26.33	1800.39	77.65	0	156.52	77.65	24.38	1800.78
I5_R10_M2-5	68.60	29.83	1800.40	67.63	0	500.74	67.63	40.06	1801.09
I5_R10_M3-1	54.46	22.52	1800.56	54.46	0	65.63	54.46	30.99	1800.61
I5_R10_M3-2	177.86	61.12	1800.60	177.86	0	31.74	177.86	0	945.12
I5_R10_M3-3	144.62	48.30	1800.72	144.62	0	3.89	144.62	0	315.00
I5_R10_M3-4	82.47	33.57	1800.60	82.47	0	154.40	82.47	15.43	1800.41
I5_R10_M3-5	104.16	53.04	1800.81	104.16	0	6.27	104.16	0	222.59
I5_R20_M2-1	143.92	51.59	1801.38	143.92	50.36	1800.92	143.92	51.90	1800.84
I5_R20_M2-2	129.28	54.04	1801.74	128.46	52.27	1800.89	129.28	60.74	1801.43
I5_R20_M2-3	133.14	60.41	1801.78	124.82	56.39	1800.94	125.00	88.33	1802.12
I5_R20_M2-4	121.16	41.39	1801.69	121.16	40.14	1800.80	121.16	40.14	1801.90
I5_R20_M2-5	150.08	46.37	1801.59	143.35	42.45	1800.80	143.35	42.55	1801.02

I5_R20_M3-1	114.46	59.16	1802.23	104.63	53.89	1800.95	105.60	98.70	1802.57
I5_R20_M3-2	160.33	61.40	1802.41	160.33	60.37	1801.10	160.33	60.75	1802.00
I5_R20_M3-3	135.47	56.42	1802.16	139.39	56.29	1800.93	139.39	64.94	1801.88
I5_R20_M3-4	142.55	43.76	1802.12	136.80	39.81	1800.99	136.80	39.81	1802.04
I5_R20_M3-5	139.30	61.15	1802.18	134.08	58.32	1801.09	134.08	96.59	1801.72

Table 2 Detailed results of three MIPV models

Instances	MIP1V			MIP2V			MIP3V		
	Obj	Gap	Time	Obj	Gap	Time	Obj	Gap	Time
I2_R10_M2-1	154.92	0	6.92	154.92	0	0.17	154.92	0	0.13
I2_R10_M2-2	186.93	0	2.27	186.93	0	0.16	186.93	0	0.41
I2_R10_M2-3	287.88	0	8.91	287.88	0	0.07	287.88	0	0.26
I2_R10_M2-4	203.31	0	3.50	203.31	0	0.08	203.31	0	0.21
I2_R10_M2-5	196.44	0	1.44	196.44	0	0.06	196.44	0	0.13
I2_R10_M3-1	309.66	0	5.87	309.66	0	1.62	309.66	0	2.44
I2_R10_M3-2	418.75	0	2.36	418.75	0	0.53	418.75	0	0.16
I2_R10_M3-3	373.08	0	6.54	373.08	0	0.34	373.08	0	0.36
I2_R10_M3-4	260.23	0	12.09	260.23	0	1.91	260.23	0	7.49
I2_R10_M3-5	299.98	0	1.18	299.98	0	0.27	299.98	0	10.25
I2_R20_M2-1	230.00	1.20	1800.27	230.00	1.20	1801.80	230.00	1.20	1800.38
I2_R20_M2-2	271.35	8.76	1800.22	271.35	0.89	1800.33	273.44	14.11	1800.66
I2_R20_M2-3	372.45	5.55	1800.92	372.45	5.55	1801.08	372.45	5.55	1800.97
I2_R20_M2-4	271.41	18.42	1800.34	271.41	18.42	1800.99	274.84	19.44	1800.43

I2_R20_M2-5	293.65	14.43	1800.56	293.65	14.43	1800.81	294.41	14.65	1800.66
I2_R20_M3-1	284.42	9.62	1800.60	284.42	10.87	1800.75	284.42	10.99	1801.22
I2_R20_M3-2	306.60	1.49	1800.68	302.93	0.30	1804.66	303.47	0.47	1801.06
I2_R20_M3-3	325.97	16.30	1800.34	318.31	14.29	1800.39	326.85	16.53	1800.88
I2_R20_M3-4	337.40	5.32	1800.29	335.62	4.82	1800.33	334.57	4.52	1800.72
I2_R20_M3-5	345.26	36.64	1800.33	345.26	20.31	1800.51	345.26	30.92	1800.90
I5_R10_M2-1	72.69	0	0.72	72.69	0	0.22	72.69	0	0.24
I5_R10_M2-2	84.86	0	2.31	84.86	0	1.40	84.86	0	3.69
I5_R10_M2-3	56.98	0	3.62	56.98	0	1.34	56.98	0	2.70
I5_R10_M2-4	77.65	0	2.02	77.65	0	0.87	77.65	0	2.38
I5_R10_M2-5	67.63	0	3.87	67.63	0	1.09	67.63	0	3.47
I5_R10_M3-1	54.46	0	3.32	54.46	0	1.06	54.46	0	1.02
I5_R10_M3-2	177.86	0	1.23	177.86	0	0.29	177.86	0	0.22
I5_R10_M3-3	144.62	0	1.41	144.62	0	0.56	144.62	0	0.22
I5_R10_M3-4	82.47	0	2.82	82.47	0	1.16	82.47	0	0.43
I5_R10_M3-5	104.16	0	1.13	104.16	0	0.25	104.16	0	0.28
I5_R20_M2-1	143.92	22.83	1800.31	143.92	1.28	1800.15	144.49	1.66	1801.21
I5_R20_M2-2	129.28	18.28	1800.44	128.46	13.11	1800.37	128.46	13.11	1800.81
I5_R20_M2-3	125.23	2.05	1800.49	124.82	1.72	1800.47	125.26	2.07	1801.09
I5_R20_M2-4	121.16	34.99	1800.38	121.16	0	1.58	121.16	0	8.65
I5_R20_M2-5	143.35	16.02	1800.46	143.35	0	2.25	143.35	16.02	1800.75
I5_R20_M3-1	104.63	8.19	1800.62	105.89	9.28	1800.17	107.33	10.50	1801.51
I5_R20_M3-2	160.33	37.38	1800.57	160.33	0	39.51	160.33	0	22.57

I5_R20_M3-3	137.23	1.70	1800.85	134.90	0	2.68	137.28	1.74	1801.34
I5_R20_M3-4	136.80	14.89	1800.73	136.80	0	4.64	136.80	0	131.08
I5_R20_M3-5	134.08	18.24	1800.47	134.08	1.90	1800.21	134.08	18.24	1801.27

Table 3 Detailed results of MIP2V and LBBD of small-sized instances

Instances	MIP2V			LBBD		
	Obj	Gap	Time	Obj	Gap	Time
I2_R10_M2-1	154.92	0	0.17	154.92	0	0.10
I2_R10_M2-2	186.93	0	0.16	186.93	0	0.22
I2_R10_M2-3	287.88	0	0.07	287.88	0	0.08
I2_R10_M2-4	203.31	0	0.08	203.31	0	0.05
I2_R10_M2-5	196.44	0	0.06	196.44	0	0.09
I2_R10_M3-1	309.66	0	0.53	309.66	0	0.35
I2_R10_M3-2	418.75	0	0.34	418.75	0	0.10
I2_R10_M3-3	373.08	0	1.62	373.08	0	0.08
I2_R10_M3-4	260.23	0	1.91	260.23	0	0.31
I2_R10_M3-5	299.98	0	0.27	299.98	0	0.09
I2_R20_M2-1	230.00	1.20	1801.80	230.00	0	80.54
I2_R20_M2-2	271.35	0.89	1800.33	271.35	0	225.12
I2_R20_M2-3	372.45	5.55	1801.08	372.45	0	732.83
I2_R20_M2-4	271.41	18.42	1800.99	271.41	0	1069.38
I2_R20_M2-5	293.65	14.43	1800.81	293.65	0	1093.30
I2_R20_M3-1	284.42	10.87	1800.75	284.23	0	1640.08

I2_R20_M3-2	302.93	0.30	1804.66	302.93	0.30	1800.23
I2_R20_M3-3	318.31	14.29	1800.39	318.31	0	105.18
I2_R20_M3-4	335.62	4.82	1800.33	334.57	4.52	1800.13
I2_R20_M3-5	345.26	20.31	1800.51	345.26	0	89.12
I3_R10_M2-1	115.64	0	0.30	115.64	0	0.22
I3_R10_M2-2	133.61	0	0.15	133.61	0	0.08
I3_R10_M2-3	152.90	0	0.10	152.90	0	0.06
I3_R10_M2-4	120.81	0	0.58	120.81	0	0.22
I3_R10_M2-5	111.68	0	0.97	111.68	0	0.26
I3_R10_M3-1	151.67	0	0.14	151.67	0	0.05
I3_R10_M3-2	142.11	0	3.99	142.11	0	0.36
I3_R10_M3-3	89.07	0	2.65	89.07	0	0.21
I3_R10_M3-4	151.80	0	0.18	151.80	0	0.12
I3_R10_M3-5	127.91	0	0.38	127.91	0	0.08
I3_R20_M2-1	222.68	10.91	1800.13	222.68	0	141.02
I3_R20_M2-2	182.61	9.38	1801.61	182.61	0	265.02
I3_R20_M2-3	214.71	18.45	1800.45	214.02	12.37	1800.10
I3_R20_M2-4	178.04	0	41.76	178.04	0	5.48
I3_R20_M2-5	209.60	11.00	1800.32	209.60	0	378.28
I3_R20_M3-1	195.09	13.15	1800.13	195.09	0	108.99
I3_R20_M3-2	207.13	20.27	1801.70	206.89	0	582.76
I3_R20_M3-3	426.39	0	111.38	426.39	0	0.74
I3_R20_M3-4	244.43	13.73	1800.15	244.43	0	295.14

I3_R20_M3-5	494.81	6.50	1800.84	494.81	0	4.74
I5_R10_M2-1	72.69	0	0.22	72.69	0	0.12
I5_R10_M2-2	84.86	0	1.40	84.86	0	0.37
I5_R10_M2-3	56.98	0	1.34	56.98	0	0.28
I5_R10_M2-4	77.65	0	0.87	77.65	0	0.27
I5_R10_M2-5	67.63	0	1.09	67.63	0	0.39
I5_R10_M3-1	54.46	0	1.06	54.46	0	0.17
I5_R10_M3-2	177.86	0	0.29	177.86	0	0.13
I5_R10_M3-3	144.62	0	0.56	144.62	0	0.07
I5_R10_M3-4	82.47	0	1.16	82.47	0	0.19
I5_R10_M3-5	104.16	0	0.25	104.16	0	0.08
I5_R20_M2-1	143.92	1.28	1800.15	143.92	0	2.12
I5_R20_M2-2	128.46	13.11	1800.37	128.46	0	4.36
I5_R20_M2-3	124.82	1.72	1800.47	122.67	0	3.69
I5_R20_M2-4	121.16	0	1.58	121.16	0	0.39
I5_R20_M2-5	143.35	0	2.25	143.35	0	1.39
I5_R20_M3-1	105.89	9.28	1800.17	104.63	0	40.82
I5_R20_M3-2	160.33	0	39.51	160.33	0	1.58
I5_R20_M3-3	134.90	0	2.68	134.90	0	4.16
I5_R20_M3-4	136.80	0	4.64	136.80	0	3.55
I5_R20_M3-5	134.08	1.90	1800.21	134.08	0	2.40

Table 4 Detailed results of MIP2V and LBBD of medium-sized instances

	MIP2V			LBBD		
	Obj	Gap	Time	Obj	Gap	Time
I2_R30_M3-1	488.39	3.01	1800.16	487.50	2.83	1800.11
I2_R30_M3-2	422.08	0.43	1801.03	420.30	0	771.08
I2_R30_M3-3	598.98	11.86	1800.47	598.15	0.44	1800.38
I2_R30_M3-4	546.98	19.16	1800.94	546.98	0	1014.60
I2_R30_M3-5	556.91	19.19	1800.65	556.65	4.34	1800.18
I2_R30_M5-1	491.62	11.73	1800.25	490.16	11.46	1800.27
I2_R30_M5-2	457.27	0	1802.92	457.27	0	5.42
I2_R30_M5-3	516.36	4.89	20.52	513.97	4.45	1800.34
I2_R30_M5-4	423.72	0	1800.47	423.72	0	326.80
I2_R30_M5-5	481.83	14.54	395.61	479.69	14.16	1800.26
I2_R40_M3-1	736.51	19.03	1800.24	734.18	5.89	1800.28
I2_R40_M3-2	543.03	0	5.03	543.03	0	9.51
I2_R40_M3-3	715.53	0.35	1800.31	713.04	0	67.44
I2_R40_M3-4	594.37	6.61	1801.58	598.63	7.28	1800.38
I2_R40_M3-5	632.98	1.70	1801.24	630.33	1.29	1801.15
I2_R40_M5-1	657.29	12.87	1800.28	656.82	12.81	1800.31
I2_R40_M5-2	915.81	29.72	1805.18	914.42	6.47	1802.35
I2_R40_M5-3	754.29	8.35	1801.87	752.80	8.17	1807.82
I2_R40_M5-4	599.21	0.59	1801.87	599.05	1.72	1801.08
I2_R40_M5-5	673.00	15.90	1800.45	671.41	16.21	1800.67

I3_R30_M3-1	474.44	5.50	1800.69	474.44	0.15	1800.25
I3_R30_M3-2	387.72	12.50	1801.34	387.72	0	8.34
I3_R30_M3-3	300.40	0	141.28	300.40	0	8.80
I3_R30_M3-4	489.10	5.82	1800.40	488.66	0	7.58
I3_R30_M3-5	352.79	11.95	1801.00	349.43	10.89	1800.16
I3_R30_M5-1	319.89	23.32	1800.70	319.62	23.26	1800.30
I3_R30_M5-2	346.73	9.97	1800.95	339.87	8.15	1801.95
I3_R30_M5-3	321.51	7.41	1800.99	313.97	5.19	1800.89
I3_R30_M5-4	376.27	9.39	1800.72	371.75	8.29	1801.26
I3_R30_M5-5	348.15	15.89	1800.63	347.81	15.81	1800.98
I3_R40_M3-1	467.21	11.82	1800.98	466.37	11.66	1800.24
I3_R40_M3-2	373.63	2.17	1800.97	371.37	1.58	1801.54
I3_R40_M3-3	636.43	18.85	1800.91	636.02	0	282.66
I3_R40_M3-4	408.70	0	446.25	408.70	0	31.92
I3_R40_M3-5	443.98	5.96	1800.82	443.16	5.24	1800.25
I3_R40_M5-1	384.36	7.13	1801.17	381.73	6.96	1800.41
I3_R40_M5-2	367.74	16.04	1801.30	366.99	15.87	1801.33
I3_R40_M5-3	471.19	7.30	1801.59	465.30	6.13	1801.81
I3_R40_M5-4	386.86	6.14	1801.18	383.25	5.25	1801.46
I3_R40_M5-5	458.12	6.90	1801.12	450.59	4.89	1801.19
I5_R30_M3-1	264.88	0	76.87	264.88	0	2.44
I5_R30_M3-2	192.95	13.64	1800.86	186.26	10.54	1800.27
I5_R30_M3-3	206.90	24.88	1800.92	206.69	14.45	1800.79

I5_R30_M3-4	190.04	17.26	1800.96	189.18	11.79	1801.98
I5_R30_M3-5	629.02	9.49	1800.76	627.64	0	12.77
I5_R30_M5-1	203.53	15.86	1801.08	198.12	4.62	1800.36
I5_R30_M5-2	590.15	5.82	1801.13	589.26	0	22.13
I5_R30_M5-3	173.53	8.98	1801.23	172.07	8.21	1800.77
I5_R30_M5-4	308.71	8.53	1801.26	308.71	0	7.76
I5_R30_M5-5	205.37	10.34	1801.14	196.27	8.46	1800.33
I5_R40_M3-1	227.64	12.14	1801.26	223.77	10.62	1800.37
I5_R40_M3-2	281.09	15.16	1801.12	270.98	9.26	1800.95
I5_R40_M3-3	304.49	7.45	1801.35	306.40	8.03	1802.04
I5_R40_M3-4	233.22	9.14	1801.15	231.98	8.65	1800.90
I5_R40_M3-5	225.00	0	351.74	225.00	0	42.66
I5_R40_M5-1	307.25	22.55	1801.66	307.01	0	14.56
I5_R40_M5-2	217.50	23.77	1801.87	217.50	22.96	1800.53
I5_R40_M5-3	277.53	22.09	1801.64	277.53	0	20.68
I5_R40_M5-4	316.87	21.37	1801.50	316.87	0	6.67
I5_R40_M5-5	246.92	7.80	1801.88	243.25	6.41	1800.45

Table 5 Detailed results of MIP2V and LBBD of large-sized instances

	MIP2V			**LBBD**		
	Obj	Gap	Time	Obj	Gap	Time
I5_R50_M5-1	300.47	20.41	1800.99	296.84	19.44	1800.68
I5_R50_M5-2	318.23	3.52	1802.18	312.82	1.85	1801.29

I5_R50_M5-3	315.39	9.00	1802.65	314.94	0	1085.59
I5_R50_M5-4	450.87	7.78	1801.64	450.87	1.84	1801.56
I5_R50_M5-5	320.18	4.67	1803.28	314.66	3.00	1803.57
I5_R50_M8-1	335.60	11.29	1803.12	324.70	8.31	1800.97
I5_R50_M8-2	423.07	22.99	1807.26	423.07	0	182.91
I5_R50_M8-3	476.05	32.75	1803.11	475.10	0.52	1801.37
I5_R50_M8-4	572.47	6.87	1803.56	571.30	1.60	1803.58
I5_R50_M8-5	539.01	8.69	1802.97	537.43	3.46	1802.36
I5_R80_M5-1	578.21	23.22	1802.54	545.13	6.47	1801.76
I5_R80_M5-2	527.07	1.80	1802.28	517.57	0	158.49
I5_R80_M5-3	531.55	9.86	1801.93	516.12	7.17	1801.87
I5_R80_M5-4	487.75	0	798.27	487.75	0	317.75
I5_R80_M5-5	490.10	0	1031.75	490.10	0	179.93
I5_R80_M8-1	534.59	15.85	1800.10	522.89	13.97	1802.54
I5_R80_M8-2	522.09	4.31	1801.57	499.59	0	544.38
I5_R80_M8-3	589.10	16.31	1802.79	581.51	15.22	1808.51
I5_R80_M8-4	579.14	10.61	1806.90	570.34	9.23	1811.18
I5_R80_M8-5	910.22	45.36	1803.37	909.04	4.10	1812.68
I5_R100_M5-1	931.41	34.64	1803.28	927.05	3.04	1804.35
I5_R100_M5-2	697.79	9.69	1804.02	666.76	5.48	1808.71
I5_R100_M5-3	1193.08	48.18	1807.26	1185.77	7.59	1807.96
I5_R100_M5-4	757.78	24.74	1807.88	756.27	24.59	1803.84
I5_R100_M5-5	835.24	34.59	1803.58	830.78	20.72	1804.38

I5_R100_M8-1	1013.45	42.44	1803.90	1004.79	6.69	1804.57
I5_R100_M8-2	727.98	13.09	1804.15	695.18	8.99	1814.71
I5_R100_M8-3	750.86	20.78	1804.58	656.02	9.33	1805.17
I5_R100_M8-4	660.84	12.47	1804.68	617.09	6.26	1806.84
I5_R100_M8-5	1779.24	61.40	1803.75	1775.30	19.31	1810.59
I10_R50_M5-1	154.06	16.68	1801.72	140.76	8.80	1801.44
I10_R50_M5-2	165.89	12.62	1801.75	156.32	7.28	1801.81
I10_R50_M5-3	159.29	6.21	1801.86	159.29	0	97.13
I10_R50_M5-4	148.43	6.17	1803.02	145.99	4.60	1801.50
I10_R50_M5-5	188.00	10.62	1804.99	188.00	0	69.79
I10_R50_M8-1	177.87	19.82	1802.40	165.25	13.69	1801.93
I10_R50_M8-2	195.12	20.88	1802.59	195.12	0	23.86
I10_R50_M8-3	175.70	12.82	1802.88	166.79	8.16	1801.93
I10_R50_M8-4	178.45	13.62	1803.36	178.45	0	20.76
I10_R50_M8-5	450.36	17.15	1806.14	449.89	0	55.59
I10_R80_M5-1	255.96	15.20	1803.60	236.54	8.23	1803.25
I10_R80_M5-2	839.41	6.37	1804.00	838.43	5.13	1810.06
I10_R80_M5-3	285.05	16.30	1803.94	276.49	13.71	1805.96
I10_R80_M5-4	254.91	13.62	1804.39	246.88	10.81	1804.47
I10_R80_M5-5	847.83	70.87	1810.67	847.28	8.85	1813.65
I10_R80_M8-1	255.93	14.16	1805.28	239.79	8.38	1804.80
I10_R80_M8-2	293.17	15.65	1805.86	255.93	3.37	1805.22
I10_R80_M8-3	329.61	31.37	1813.88	296.87	7.37	1804.81

I10_R80_M8-4	294.22	16.00	1806.06	289.79	14.72	1805.99
I10_R80_M8-5	489.43	49.82	1812.67	488.70	0	378.53
I10_R100_M5-1	384.08	26.08	1806.30	351.05	10.88	1805.44
I10_R100_M5-2	437.50	25.25	1805.69	383.73	14.77	1806.03
I10_R100_M5-3	476.84	37.47	1805.66	386.62	9.31	1814.12
I10_R100_M5-4	452.60	37.75	1807.30	429.42	0	621.66
I10_R100_M5-5	392.66	24.61	1805.41	370.21	10.34	1805.19
I10_R100_M8-1	434.34	35.42	1807.23	427.33	5.47	1806.84
I10_R100_M8-2	550.15	44.20	1801.31	366.30	7.63	1808.74
I10_R100_M8-3	480.01	42.34	1807.22	692.36	0	485.98
I10_R100_M8-4	495.36	31.69	1811.16	406.14	0.93	1806.58
I10_R100_M8-5	696.36	57.84	1801.26	335.81	16.47	1806.91